09C — Deep-Dive Feature Audit: Combat Engine & Tactical Simulation (Part 3 of 6)

Turn/Tick Action Economy, Charged/Channeled Multiplier Loss, The `statusTick` DoT Self-Harm & Cross-Map Reflect Cascade, Shield Overwrites, Acupoint Bugs & Boss Anatomy Breaker

- **Target Codebase**: `JianghuSect 2` (`JS2-v2.6.0` in Google Drive `Js2 new`)
- **Target Audience**: Autonomous AI Developers (**GPT** & **Gemini**)
- **Core Game Philosophy**: **100% Emergent, Causally Sound Combat Mechanics**. Poison ticking inside an enemy's veins must never shoot sword-beams back across the map or drain the poisoner's own blood; charging an ultimate must preserve elemental and mastery multipliers; and acupuncture/part-breaking must behave according to Wuxia martial logic!
- **Master Roadmap for Category 09 (`09A–09F`)**:
  - `09A`: Core Damage Formula (`dmg = hpMax(d) * frac`), 12-Stat Split-Brain, Elemental Matchups & Behavioral Mastery Traits
  - `09B`: Spatial Battlefield Geometry, `NPC` vs `foe` Movement Asymmetry, `24`-Tile `ROW_BODYBLOCK` Bug, Interception & AoE Selectors
  - **`09C` (This Document)**: **Action Economy, Charged/Channeling Multiplier Loss, `statusTick` DoT Self-Harm & Reflect Loops, Shield Overwrite, `BLOOD_HALT` Acupoint Bug & Boss `ANATOMY_BREAKER`**
  - `09D`: Multi-Entity Units — Summons, Shadow Clones, Constructs, Pet Assists, Gate Beasts, Weapon Avatars, Ancestral Phantoms & Party Synergies
  - `09E`: Fight Modes, Stakes, Morale/Yield/Retreat, 7-Layer Cheat-Death Stacking, Raid Core-Breach AI, Self-Defense Jail Bug & Heavenly Tribulations
  - `09F`: Grand Finale — Holistic Overview of the Entire Combat Ecosystem

---

1. `[CMB-DOT-01]` Critical Cascade in `C.statusTick` $\to$ `C.applyDamage` (`js/combat.js:1005–1011` $\to$ `265–385`): Every 1-Minute Poison/Bleed Tick Triggers Direct-Strike Procs, Cross-Map Reflects, Free Taiji Qi & Self-Blood-Drain!

Inspect `C.statusTick` (`js/combat.js:1005–1011`):

```javascript
if (S.dot && x.per) {
  const src = x.src ? C.unit(W, x.src) : null;
  ...
  C.applyDamage(W, f, src && !src.dead ? src : null, u, C.hpMax(u) * x.per * ...);
}
```

Because `C.statusTick` calls `C.applyDamage` with `a = src` and **`note === undefined`** (no `'DOT'` / `opts.isDot` guard), `C.applyDamage` treats every internal DoT tick identically to a fresh melee weapon swing:

1. **Poisoner Bleeds Themselves to Death (`lines 343–347`)**: If `src` has an unsealed Forbidden Art (`fb.isUnsealed`), every DoT tick on `6` poisoned foes deducts `hpCost = 3.5% MaxHP` from `src` (`-21% MaxHP/min` self-harm from their own poison!) and spikes `tabooStrain`!
2. **Cross-Map Laser Reflection & Free Qi Battery (`lines 284–294, 317–328`)**: If poisoned victim `u` has `TAIJI_SOFT_REDIRECTION`, `VAJRA_IMMORTAL_BODY_BELL`, or `SWORD_QI_MERIDIAN`, every tick of internal poison **restores `+5 Qi` to `u` AND fires `12%–22%` reflected damage `25` tiles across the map back into `src`**!
3. **Uncapped Evergreen Multi-DoT Healing (`lines 274–276`)**: `EVERGREEN_LONGEVITY_SPRING` heals `src` `+3.5% MaxHP` on every single DoT tick across every poisoned enemy!
4. **Posthumous DoT Kills Erase Killer Attribution (`line 1011`)**: `src && !src.dead ? src : null`—if `src` dies 1 tick before their poison kills `u`, `a` becomes `null`, so `C.slay` records `killer = null` (losing kill credit, `DEMON` soul-harvest, weapon kill count, and `BLACK_THREAD` karma!).

- **Exact Fix**: Pass `note = 'DOT:' + x.k` (and keep `src` even if `src.dead` purely for `C.slay` attribution while guarding `!a.dead` on heals/counters!), and skip direct-hit Mantra/Defensive/Forbidden strike hooks inside `C.applyDamage` whenever `String(note || '').startsWith('DOT:')`!

---

2. `[CMB-CHAN-02]` Four Bugs in Charged Skills, Channeling & `TECH_GCD` (`js/combat.js:613–618, 738–740, 767–771, 1044–1055, 1152–1175`)!

1. **`C.resolveChannelTick` Drops `100%` of `synDmgMult`, `elemDmgMult`, `masBase` & `wpnAcc` (`lines 1046–1052`)**:

- `useSkill` (`lines 732–742`) carefully computes `p.synDmgMult = (syn ? syn.dmgMult : 1) * elemDmgMult` and `masBase = SK.masteryBase(a, p)`.
- Yet `C.resolveChannelTick` (`line 1046`) calls `C.strike(W, f, a, t, { M: mult, accAdd: (ch.skill && ch.skill.elementAccAdd) || 0, type: ..., pen: ... })` **WITHOUT passing `synDmgMult: ch.skill?.synDmgMult` and WITHOUT multiplying `masBase` or `wpnAcc`**! Charging a Grandmaster Elemental Ultimate loses up to `1.35 × 1.22 × 1.40 = 2.30×` of its rightful scaling!

2. **`startChargedSkill` (`line 1168`) Deals Premature `tickM: 0.35` Poke Strikes While "Charging"**:
   - Unlike multi-pulse `channel` beams, a *Charged Slash (蓄力斩, `isChargedSkill === true`)* gathers Qi on non-final ticks (`ticksLeft > 1`) before unleashing `finishM` (`1.65×–1.85×`) on the final tick (`verified compatible with v340:157–160, which only asserts caster.lastChargedRelease.releaseMult === 1.65 after 2 ticks!`). Currently, Tick 1 pokes for `0.35×`, prematurely triggering enemy `counter` / `astralMirage` ripostes! On `ch.isChargedSkill && !isFinal`, gather Qi (`fx: '⚡ รวบรวมปราณ...'`) without firing a premature poke!
3. **`TECH_GCD` Skipped on Move 1 (`line 739`: `if (p.tech && p.tech !== p.id)`)**:
   - Always key technique-wide GCD on `const techKey = p.tech || p.id` whenever a parent technique has ≥ 2 unlocked moves so Move 1 $\to$ Move 2 obeys the same cadence as Move 2 $\to$ Move 1!
4. **`C.leave` (`line 696`) Wipes `vowBacklash` & Forgets `_poiseGauge` / `_ccDrStacks`**:
   - Preserve persistent curse/vow statuses (`x.persistAfterCombat || x.k === 'vowBacklash'`) in `C.leave`, while resetting combat-only `n._poiseGauge = 100; n._ccDrStacks = 0; n._ccImmuneUntil = 0;`!

---

3. `[CMB-ACU-03]` Broken `'BLOOD_HALT'` Acupoint Allies-Harm Bug (`js/combat.js:1084–1104`), Shield Overwrite Downgrade, & Boss `POISE_BREAK` Stunlock!

1. **Acupoint `'BLOOD_HALT'` Inflicts `healBlock` on Friendly Allies Instead of Healing Them (`lines 1084–1104`)!**:
   - Inspect lines `1084–1104` verbatim:

```
JavaScript
BLOOD_HALT: { id: 'BLOOD_HALT', ..., status: 'BLOOD_HALT', secondaryStatus:
'healBlock', allyHealPct: 0.18,
  descTh: 'ลดความเร็วเคลื่อนที่และระงับการฟื้นฟูเลือดของศัตรู หรือใช้ห้ามเลือดฟื้นฟูพันธมิตร' },
...
if (spec.id === 'BLOOD_HALTING_POINT' && sameSide) { // <-- FORGOT || spec.id ===
'BLOOD_HALT'!!
```

  - **Two Bugs**:
    - Against an **ally** (`sameSide === true`), `'BLOOD_HALT'` skips line `1096` and executes lines `1103–1104`, **inflicting `healBlock` (`-75%` healing!) on your own dying teammate instead of healing `+18%` HP**!
    - Against an **enemy**, `spec.status: 'BLOOD_HALT'` is an **unregistered uppercase key** (`ignored by SK.STATUS and stats.js`, so `"ลดความเร็วเคลื่อนที่"` never applies—change `status` to `'slow'`)!
2. **Small Proc Shields Overwrite Massive Active Shields (`ST.addStatus(u, 'shield', ...)`)**:
  - When a unit holds a `35% MaxHP` Tetra-Reaction / Barrier Shield (`amt = 1,400`) and triggers a minor `10%` proc shield (`amt = 400`), overwriting `amt` destroys `1,000` shield HP! In `ST.addStatus` for `k === 'shield'`, set `amt = Math.max(cur.amt || 0, extra.amt || 0)` (and `until = Math.max(cur.until, nextUntil)`).
3. **`POISE_BREAK` (`lines 1124–1131`) Ignores `_ccImmuneUntil` DR Window**:
  - Check `!(t._ccImmuneUntil && tNow < t._ccImmuneUntil)` before applying the 4-tick `stun` on `POISE_BREAK` (still applying `poiseBroken` `+35%` damage vulnerability!) so multi-hit parties cannot permanently stunlock Tier-7 Bosses!

---

4. `[CMB-ANAT-04]` Tactical & Catalog Gaps in Boss Beast `ANATOMY_BREAKER` (`js/combat.js:569–575, 1179–1205`)!

1. **Preserve `v340:176` Direct-Call Contract While Preventing Mid-Defeat Loot Teleportation**:
  - Note: `test/v340_combat_depth_and_duet_synergies_suite.js:176` calls `damageBossPart(W, f, hunter, boss, 'VENOM_SAC', ...)` directly (`opts.fromStrike` is `undefined`) and immediately asserts `W.sect.stock.MAT_BOSS_VENOM_SAC === 1`.
  - **Test-Safe Fix**: Keep immediate `W.sect.stock` deposit when `!opts.fromStrike` (preserving `v340:176` 100%!), while during live `C.strike` (`opts.fromStrike === true`), store broken organ drops in

`f.escrowBossDrops` and commit them to `hunter.inv.mats` / `W.sect.stock` inside `C.conclude` **only when `f.winner === 'A'`** (so hunters who flee or wipe mid-fight don't teleport organs home)!

2. **Register All `4` `MAT_BOSS_*` Organ Items in `ECO.matById` & Tactical Part Selection (`line 571`)**:
    - Ensure `MAT_BOSS_HORN_CROWN`, `MAT_BOSS_VENOM_SAC`, `MAT_BOSS_WING`, and `MAT_BOSS_CORE_SHELL` exist in `data_items.js` / `ECO.matById` (`tier: 5, cat: 'BEAST_PART', value: 420`) and work in `CONTENT.rareOrgans` / Tier-5 alchemy & forging!
    - Replace naive `Object.keys(parts).find` (`always hitting HORN_CROWN first`) with **smart tactical targeting**: `GRAND_TACTICIAN` / `HEALER` targets `VENOM_SAC` first against venomous bosses; `RANGED_KITE` targets `WING_MEMBRANE`; `HEAVY_BREAKER` targets `CORE_SHELL`!

---

5. Mandatory Verification Suite Additions (`09C`)

1. **DoT Isolation Test**: Inflict `poison` from an unsealed Forbidden-Art caster onto a `TAIJI_SOFT_REDIRECTION` defender at `dist = 15`; tick `C.statusTick` 5 times and assert `0` self-HP drain on the poisoner, `0` cross-map reflect damage, and accurate `killer` credit if the poisoner dies 1 tick before the victim!
2. **Charged-Skill Full Multiplier & `'BLOOD_HALT'` Ally-Heal Test**: Verify `resolveChannelTick` applies `synDmgMult * masBase` and `executeAcupointStrike(..., ally, 'BLOOD_HALT')` heals `+18%` without applying `healBlock`.
3. **Shield Non-Downgrade Test**: Apply a `500`-HP shield followed by a `150`-HP proc shield and assert `shieldOf(u).amt === 500`.